Project 2: Mathematical Formulas

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Realized variance

$$\bar{f}_t = \frac{1}{n_t} \sum_{d \in \mathcal{D}_t} f_d, \qquad RV_t = \sum_{d \in \mathcal{D}_t} \left(f_d - \bar{f}_t\right)^2 \tag{1}$$

Portfolio formation

$$w_{t+1} = \frac{c}{RV_t}, \qquad f_{t+1}^{\mathrm{VM}} = w_{t+1} f_{t+1} \tag{2}$$

Equal-volatility normalization

$$c = \frac{s_{\mathcal{T}}(f_t)}{s_{\mathcal{T}}(f_t / RV_{t-1})}, \qquad s_{\mathcal{T}}(x) = \sqrt{\frac{1}{T-1} \sum_{t \in \mathcal{T}} (x_t - \bar{x})^2} \tag{3}$$

Univariate regression

$$f_t^{\mathrm{VM}} = \alpha_m + \beta f_t + \varepsilon_t \tag{4}$$

Annual alpha and its standard error

$$\alpha_{\mathrm{ann},\%} = 1200 \alpha_m, \qquad \mathrm{SE}(\alpha_{\mathrm{ann},\%}) = 1200\, \mathrm{SE}(\alpha_m) \tag{5}$$

Sharpe ratio and original-table RMSE units

$$\mathrm{SR}_{\mathrm{ann}} = \sqrt{12}\, \frac{\bar{f}^{\mathrm{VM}}}{s_{\mathcal{T}}(f^{\mathrm{VM}})}, \qquad \mathrm{RMSE}_{\mathrm{paper}} = 1200 \sqrt{\frac{\sum_{t \in \mathcal{T}} \hat{\varepsilon}_t^2}{T-2}} \tag{6}$$

Total-return wealth and risk

$$\begin{aligned} R_t^{\mathrm{gross}} &= r_t^f + w_t f_t, & V_t &= V_{t-1}(1 + R_t), \quad V_0 = 1, \\ \mathrm{CAGR} &= \left(\frac{V_T}{V_0}\right)^{12/T} - 1, & \mathrm{MDD} &= \max_{0 \le t \le T} \left(1 - \frac{V_t}{\max_{0 \le s \le t} V_s}\right). \end{aligned} \tag{7}$$

Leverage cap

$$w_t^{\mathrm{cap}} = \min\left\{ \frac{c}{RV_{t-1}},\, 2 \right\} \tag{8}$$

Exact proportional trading cost

$$\phi_t = k\,|w_t(1-\phi_t) - d_t| = \begin{cases} \dfrac{k(w_t - d_t)}{1 + kw_t}, & w_t \geq d_t, \\ \dfrac{k(d_t - w_t)}{1 - kw_t}, & w_t < d_t. \end{cases} \tag{9}$$

Financing and net total return

$$R_t^{\text{net}} = (1-\phi_t)\left[1 + r_t^f + w_t f_t - \frac{\lambda}{12}\max(w_t - 1, 0)\right] - 1, \qquad f_t^{\text{net}} = R_t^{\text{net}} - r_t^f \tag{10}$$